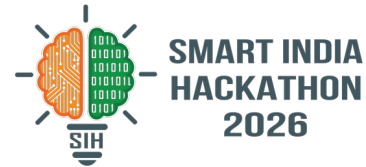
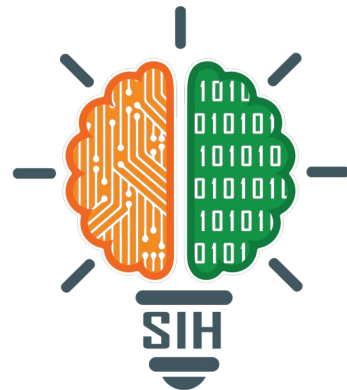


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY
- Theme – Agriculture, FoodTech & Rural Development
- PS Category – Software
- Team ID – _____ (fill from the SIH portal)
- Team Name (Registered on portal) – Cognify



SMART INDIA HACKATHON 2026

What is the problem? SIH26097 · Ministry of Social Justice & Empowerment
PM-AJAY is a central scheme for poor Scheduled Caste (SC) families. Its Grant-in-Aid (GIA) part pays for **skill training** (at least 10% of GIA funds) and **small-business support** (up to ₹50,000 or 50% of the asset cost, paid towards a bank loan). Today the course for each person is chosen through paper forms and field officers. People who cannot read, do not know the modern trades, or speak only a dialect are put in the wrong course → drop-outs, no job, money spent without result.

41%

of trained people got a job
— CAG audit of PMKVY,
2025

34%

of SC adults cannot read
— Census 2011

36%

of women have no phone
of their own — NFHS-6,
2023-24

PROPOSED SOLUTION — WHAT HAPPENS ON ONE CALL (STEPS THE PERSON EXPERIENCES)



1. Missed call → free call-back

The family gives a free missed call. Within ~1 minute SkillCall calls back from an official 1600-series government number, so the call costs the family nothing.



2. Language + basics by keypad

Press 1 Hindi, 2 Gujarati ... (22 languages possible). Consent, age group, education and travel distance are pressed on keys — a speech mistake can never change eligibility.



3. Talk about work and skills

"Aap kya kaam karte ho?" The AI maps the answer to an official NCO-2015 occupation, then asks skill questions that depend on the last answer (bike repair → engine repair? tyre change? oil service?).



4. Read back → saved as a table

Everything is read back; press 1 to confirm, 2 to correct. The confirmed profile is stored as one clean row: name, age, education, occupation, skills, distance, job-or-business.



5. Best 1-4 options, checked, explained

Profile is matched to government rules, NQR courses, training centres and local jobs. Every fact is verified by the SATYA validator before it is spoken. He can ask anything about any option.



6. Choose → CSC → officer → follow-up

Press 1-4 to choose. SMS with Case ID + nearest CSC. Documents and Aadhaar e-KYC at the CSC; the officer approves in the AJAY app. Automatic calls at 1, 3 and 6 months check he is earning.

THE 5 GIA ISSUES IN THE BRIEF → OUR ANSWER

No planning roadmap → demand map by block + draft projects for the Annual Action Plan

Identifying participants → one verified profile per person + recorded interest assessment for the selection committee

Placement after training → honest option details before joining + 1/3/6-month calls + outcome proof

Coordination gaps → one shared case record seen by CSC, officer, training partner and bank

Too few ground staff → AI does the routine calls; humans only approve and handle hard cases

INNOVATION & UNIQUENESS — 8 THINGS SKILLCALL DOES DIFFERENTLY



Everything on a call — any keypad phone, no internet, no reading — in his own regional language



Tables, not forms — AI stores every answer in structured tables; officers only review ready, arranged data + documents



Closed knowledge only — answers come only from the government data we feed it (rules, NQR, NCO, centres) — no internet search



Smart model switching — cheapest model by default; regional-language or 105B model only when needed; still unhappy → human calls



Ask anything — every doubt about a course, scheme, loan or job is answered by the LLM in his language



Adaptive questions — next question depends on past answers (bike mechanic → engine repair, tyre change, oil service)



Validator AI (SATYA) — every important fact is checked a second time against the database before it is spoken



Web portal too — login with mobile + OTP; the person, the CSC or the officer can view / edit the profile any time, or do it all on the web

TECHNOLOGIES USED (LOGO + NAME)

Calls & voice engine

 Exotel / Plivo	 WhatsApp API	 SMS (DLT)
 Pipecat	 Silero VAD	 1600 govt no.

AI models (Indian, open where possible)

 Saaras v3 - STT	 Bulbul v3 - TTS	 Sarvam-30B
 Sarvam-105B	 IndicConformer	 BGE-M3
 Bhashini (MeitY)	 SATYA (ours)	 Gemini-3 (test)

Backend & data

 Python	 FastAPI	 Celery
 Redis	 PostgreSQL	 pgvector
 PostGIS	 Pydantic	 JWT + audit log

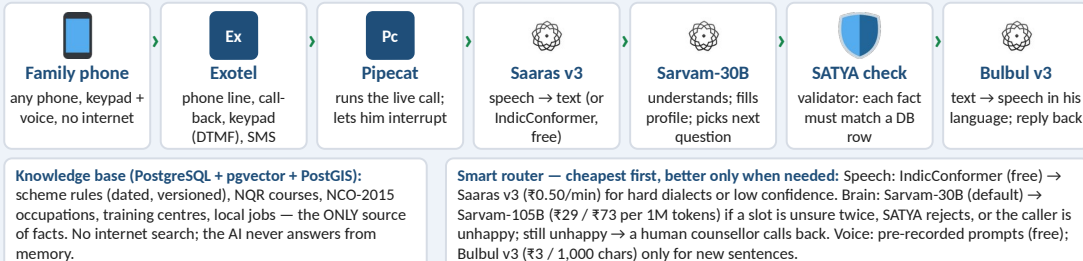
Apps, hosting & government data

 React	 Leaflet maps	 Docker
 MeghRaj cloud*	 NCO-2015	 NQR / NCVET
 DigiLocker*	 AJAY app*	 CSC / e-KYC*

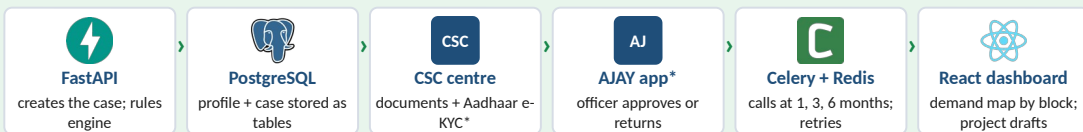
* needs government permission (MoU). Logos: Devicon / Simple Icons; Sarvam mark from Sarvam AI.

HOW IT WORKS — DURING THE CALL (LIVE) AND AFTER THE CALL

① DURING THE CALL — every sentence the person says goes round this loop in ~1.5 s



② AFTER THE CALL — the confirmed profile becomes a case that people act on



WHAT IS STORED — PostgreSQL tables (no Aadhaar number)

families • profiles • skills • options • cases • documents • followups • consent_log • audit_log + knowledge tables: schemes, rules, courses (NQR), occupations (NCO), training_centres, jobs.
Example row → Ramesh | 34 | bike repair (NCO 7231) | engine, tyre, oil service | 10 km | own business | Case #A1042.
 Raw audio deleted after confirmation; consent logged (DPDP Rules 2025); every edit in the audit log.

WEB PORTAL (React) — same database, no call needed

Login: mobile number + OTP (JWT, role-based access). Beneficiary: see and edit his profile, options and case status — or do the whole process on the web instead of a call. CSC operator: correct stored data, upload documents. Officer: approve / return cases, demand map. Always available; every change is written to the same tables and the audit log.

AI understands → verified data decides → SATYA validator checks → humans approve

Can it really be built and run? Yes — every part of SkillCall already works somewhere in India today, every model is available by API now, and one completed profile costs about ₹37–83.

FEASIBILITY — PROOF THAT EACH PIECE ALREADY WORKS

10 languages

UIDAI already runs Sarvam's voice AI for Aadhaar — on-premise, air-gapped.

PIB / MeitY, 2025

1.2 million calls / day

Kilkari (MoHFW) automated health calls: 50% answered on the 1st try, 76% by the 3rd, 99.5% by the 9th.

BMJ Global Health, 2021

≤ 20% word error

Sarvam-Audio and IndicConformer are top tier on real Indian phone calls: 36,691 speakers, 15 languages, 536 hours.

Voice of India benchmark, 2026

Prototype-ready now

Sarvam APIs, Pipecat, Exotel trial, PostgreSQL, React — all usable today. AJAY app / DigiLocker links are mocked until an MoU.

our prototype plan

COST PER COMPLETED PROFILE — VENDOR PRICES, OUR FUNNEL

Item	How it is computed	Low	High
Telephony (Exotel / Plivo)	₹0.80–1.80 / min × 18 min (10 interview + 4 follow-ups × 2) + 10% for 30-s pulses; incomplete callers spread over completed ones	₹16.5	₹37.6
Speech → text (Saaras v3)	₹30 per hour of audio = ₹0.50 per minute	₹9.4	₹9.5
Text → speech (Bulbul v3)	₹3 per 1,000 characters × 800–2,000 dynamic characters; fixed prompts are pre-recorded once (free)	₹2.4	₹6.0
LLM (Sarvam-30B / 105B)	105B price as upper limit: ₹29.28 / 1M input, ₹73.20 / 1M output tokens; 40–90K in + 3–6K out per profile	₹1.4	₹3.1
SMS (DLT)	4–6 messages × ₹0.15–0.25	₹0.6	₹1.5
Human call-back	counsellor ₹25,000 / month ≈ ₹3.2 / min; 10–20% of profiles × 5–6 min	₹2.0	₹6.0
CSC / helper fee	₹20–40 per assisted profile; 20–40% need help	₹4.0	₹16.0
Hosting, numbers	servers, database, helpline number — spread per profile	₹1.0	₹3.0
TOTAL per completed profile	funnel: 1,000 missed calls → 76% answered × 80% finish interview × 85% finish options = 517 profiles	≈ ₹37	≈ ₹83

RISKS → HOW WE HANDLE THEM

Village dialects are hard (Bhojpuri / Maithili error 4–5× Hindi) → exact facts by keypad; every answer read back; Saaras / 105B for hard dialects; human call-back last

People ignore unknown numbers (manual blocks up 208%) → they call first; call-back comes from an official 1600 number that apps may not tag or block (TRAI, Jul 2026)

Government links need permission → works standalone today; AJAY app, DigiLocker, EPFO / ESIC plug in once the MoU is signed

VIABILITY — MONEY, PEOPLE, COMPUTE

₹0.9–1.7 crore a year for one state (1 lakh profiles) = usage ₹37–83 lakh + team of 4–6 ₹40–60 lakh + hosting ₹6–20 lakh + audits/content ₹5–10 lakh. That is 0.04–0.08% of PM-AJAY's ₹2,140 crore budget (2026–27 BE), inside the ≤5% admin / MIS line (≈₹107 crore).

Pilot: 1 block, ~1,000 families, 6 months = ₹9.4–13 lakh — one-time ₹5.5–8 lakh (2 developers × 3 months, content, security audit, field tests) + running ₹3.9–4.9 lakh. SIH prototype itself: ₹5,000–15,000.

People already on the ground — 6.5 lakh+ CSCs (≈ one per gram panchayat), 90.9 lakh SHGs; PM-AJAY portal + AJAY app launched 26 May 2026 — SkillCall hands ready cases into it, it does not replace it.

Compute: APIs at pilot, self-host at state scale — a GPU costs ₹5.7–8.1 lakh / year (IndiaAI ₹65–92 / hr) and saves ≈₹6.5 per profile → break-even at 0.9–1.25 lakh profiles / year. Sarvam models are open-source (Apache 2.0) and can run on MeghRaj government cloud.

AI could say something wrong → SATYA validator: nothing is spoken unless it matches verified government data; test set of 100 tricky questions → never invent

36% of women have no phone of their own → register with help at the nearest CSC or on the web portal; add travel, safety and hostel details for women

Local job data goes stale → every record carries source + date; older than 90 days is marked "unverified" and not promised

Who gains? Poor SC families, CSC operators, the district officers who run PM-AJAY, and the Ministry that funds it. Every number below shows its working or its source.

IMPACT IN NUMBERS

41% → 70%

Today 41% of trained people get a job (CAG 2025: 23.18 of 56.14 lakh). The MSDE norm is 70% within 3 months. Gap = 29 points. Per 1,000 trainees even 50% means 90 more people working.

CAG 2025 · MSDE Common Norms

+11 points retention

Trainees given two honest job-information sessions were 11 points (18%) more likely to stay in the placed job. SkillCall gives that honest information on the call, before he chooses.

Chakravorty et al., J. Dev. Econ. 2024

~24 staff-years / year

40-min survey + form + data entry → 10-min review (30 min saved) + 4 manual follow-up calls (20 min) = 50 min × 50,000 GIA beneficiaries = 41,700 h = 5,200 days ÷ 220 = 24 staff-years.

estimate · 2,01,006 GIA beneficiaries in 2021–25

₹12.5 crore / year

If only 5% of ~50,000 GIA cases avoid a wrong match: $2,500 \times ₹50,000$ asset support = ₹12.5 crore better used — 15–34× SkillCall's usage cost for 1 lakh profiles (₹0.37–0.83 crore).

estimate

WHAT EACH PERSON GETS



Beneficiary (SC family)

free call in own language on any phone · 1–4 honest options with place, time, benefit · list of documents and how to get a missing one · SMS Case ID · follow-up till he earns · web login (mobile + OTP) to see or edit anything



CSC operator (VLE)

the case arrives pre-filled as a table · only documents + Aadhaar e-KYC to do · can correct stored data on the portal · ₹20–40 fee per assisted profile



District officer

ready, checked cases in the AJAY app · approve or return in minutes instead of a 40-min survey · demand map by block + draft projects for the Annual Action Plan



Ministry / State corporation

one record from call to job · outcomes labelled CONFIRMED (employer, EPFO*) or SELF-REPORTED · duplicates, OTP and bank IFSC checked at entry (CAG found >94% invalid bank fields in PMKVY)

BENEFITS



Social

- right course the first time
- free, in his own language, no reading needed
- women and non-readers included (CSC / web help)



Economic

- less money wasted on wrong training
- more people earning — and proved by follow-up
- cheapest-first AI keeps cost at ₹37–83 / profile



Governance

- officers get ready, checked cases
- demand map feeds the Annual Action Plan
- one shared record: CSC, officer, partner, bank



Environment

- no paper forms
- fewer trips to offices — first contact by phone
- runs on shared government cloud (MeghRaj)

Government data & scheme documents

- SIH26097 problem statement — Ministry of Social Justice & Empowerment (sih.gov.in)
- PM-AJAY GIA norms — PIB, 23 Oct 2023 (PRID 1970043); GIA beneficiaries 2021–25 — PIB, 19 Mar 2025 (PRID 2112697)
- PM-AJAY Portal & AJAY mobile app launch — PIB, 25–26 May 2026 (PRID 2265106, 2265530)
- Notes on Demands for Grants 2026–27, Demand No. 93, MoSJE — indiabudget.gov.in (₹2,140 crore)
- CAG Report No. 20 of 2025 — PMKVY audit (41% placement) — cag.gov.in
- MSDE Common Norms — 70% placement rule — msde.gov.in; NFHS-6 (2023–24) — PIB PRID 2266600; Census 2011 SC data — MoSJE Handbook
- Skill Development Programme guidelines — pmajay.dosje.gov.in

Technology

- Sarvam AI — Saaras v3 (STT), Bulbul v3 (TTS), Sarvam-30B / 105B model cards and pricing — sarvam.ai, docs.sarvam.ai, huggingface.co/sarvamai
- AI4Bharat IndicConformer — open Indian speech recognition — ai4bharat.iitm.ac.in
- BHASHINI (MeitY) — bhashini.gov.in; Pipecat open-source voice agents — pipecat.ai; Silero VAD; BGE-M3 embeddings
- Exotel / Plivo India telephony pricing (2026); WhatsApp Business API India rate guides (2026)
- PostgreSQL + pgvector + PostGIS · FastAPI · Celery · Redis · React · Leaflet · Docker; IndiaAI GPU compute pricing (2026)

Research

- Chakravorty et al., Journal of Development Economics 169 (2024) — honest job information and retention (+11 points)
- European Economic Review (2025) — DDU-GKY trainee beliefs vs reality; Banerjee & Chiplunkar (2024) — preferences and placement
- Bashingwa et al., BMJ Global Health (2021) — Kilhari: 1.2 million calls / day (PMC8327807)
- "Voice of India" Indian speech benchmark (2026) — arxiv.org/abs/2604.19151
- Medhi et al., ACM TOCHI 18(1) 2011; Patel et al., CHI 2009; Sherwani et al., ICTD 2009 — voice interfaces for low-literacy users

Standards, policy & infrastructure

- NCO-2015 occupation codes (DGE); NQR / NCVET job roles and courses — nqr.gov.in; NCS job vacancies — ncs.gov.in
- DPD Rules 2025 — consent and data protection — PIB PRID 2190014; meity.gov.in
- TRAI clarification on 1600 / 140-series government numbers — 10 Jul 2026; PTI reports on manual call blocking
- UIDAI – Sarvam voice AI deployment — PIB / MeitY 2025; Kisan e-Mitra — PIB PRID 2117392
- CSC network (6.5 lakh+) — PIB PRID 2144094; SHGs (90.9 lakh) — PIB PRID 2146870; MeghRaj government cloud